Evaluate Dose Administered Condition checks the dose administered to see if the target dose must be repeated regardless of the other evaluation rules.

Relationship to ACIP recommendations:

- Doses which were administered after the lot expiration date or which contain a condition do not need to be evaluated.
- Examples of conditions which would prevent evaluation of a vaccine dose administered range from misadministration to recalls to cold chain breaks.

The following processing model, attribute table and decision table are used to determine if dose administered can be evaluated.

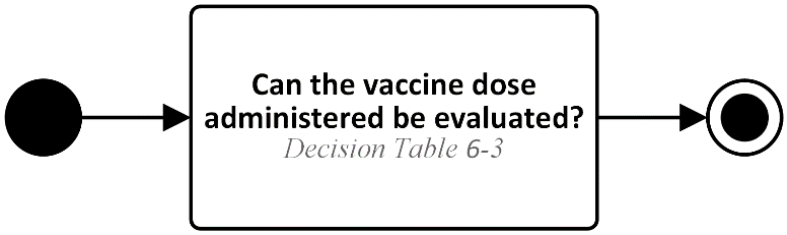

FIGURE 6-2 VACCINE DOSE ADMINISTERED CONDITION PROCESS MODEL

TABLE 6-2 DOSE ADMINISTERED CONDITION ATTRIBUTES

Attribute Type	Attribute Name	Assumed Value if Empty
Vaccine dose administered	Date Administered	-
Vaccine dose administered	Dose Condition Flag	-
Calculated date (CALCDTLOTEXP-1)	Lot Expiration Date	12/31/2999

TABLE 6-3 CAN THE VACCINE DOSE ADMINISTERED BE EVALUATED?

CONDITIONS	RULES		
Date administered > lot number expiration date?	Yes	No	No
Is the dose condition flag 'Y'?	-	Yes	No
OUTCOMES	No. The vaccine dose administered cannot be evaluated. Target dose status is 'not satisfied.' Evaluation status is 'sub-standard.'	No. The vaccine dose administered cannot be evaluated. Target dose status is 'not satisfied.' Evaluation status is 'sub-standard.'	Yes. The vaccine dose administered can be evaluated.

6.2 EVALUATE CONDITIONAL SKIP

Evaluate Conditional Skip addresses times when a target dose can be skipped. A dose should be considered necessary unless it is determined that it can be skipped. The most common scenarios for skipping a dose are:

- Catch-up doses where the patient is current with their administrations and does not need to catch-up
- The patient is behind schedule and the total number of doses needed to complete the relevant patient series can be reduced
- The previously administered dose(s) negates the need for the current target dose

Only Conditional Skip Instances with a context of Evaluation or Both should be used. In cases where a target dose does not specify Conditional Skip attributes, the target dose cannot be skipped.

A dose may be skipped based on whether or not one or more conditions evaluates to true. Conditions are classified as one of a number of types, each with one or more parameters in the Supporting Data. Conditions are contained within sets. Each set contains one or more conditions to be evaluated. Within a set, one or more conditions must be met for the set to be met. In the case where a set contains multiple conditions, whether all conditions or just one condition must be met is specified by the Condition Logic (e.g., AND vs. OR). Similarly, a dose may contain multiple sets. In the case where a dose contains multiple sets, whether all sets or just one set must be met is specified by the Set Logic.

Finally, in an effort to reduce page size and eliminate duplicate logic which could result in typographical and consistency errors, this section of logic is defined here once, but used in both Evaluation and Forecasting. The forecasting chapter refers the reader back to this section for appropriate logic.

The following process model, attribute table, and decision table are used to determine if the target dose can be skipped.

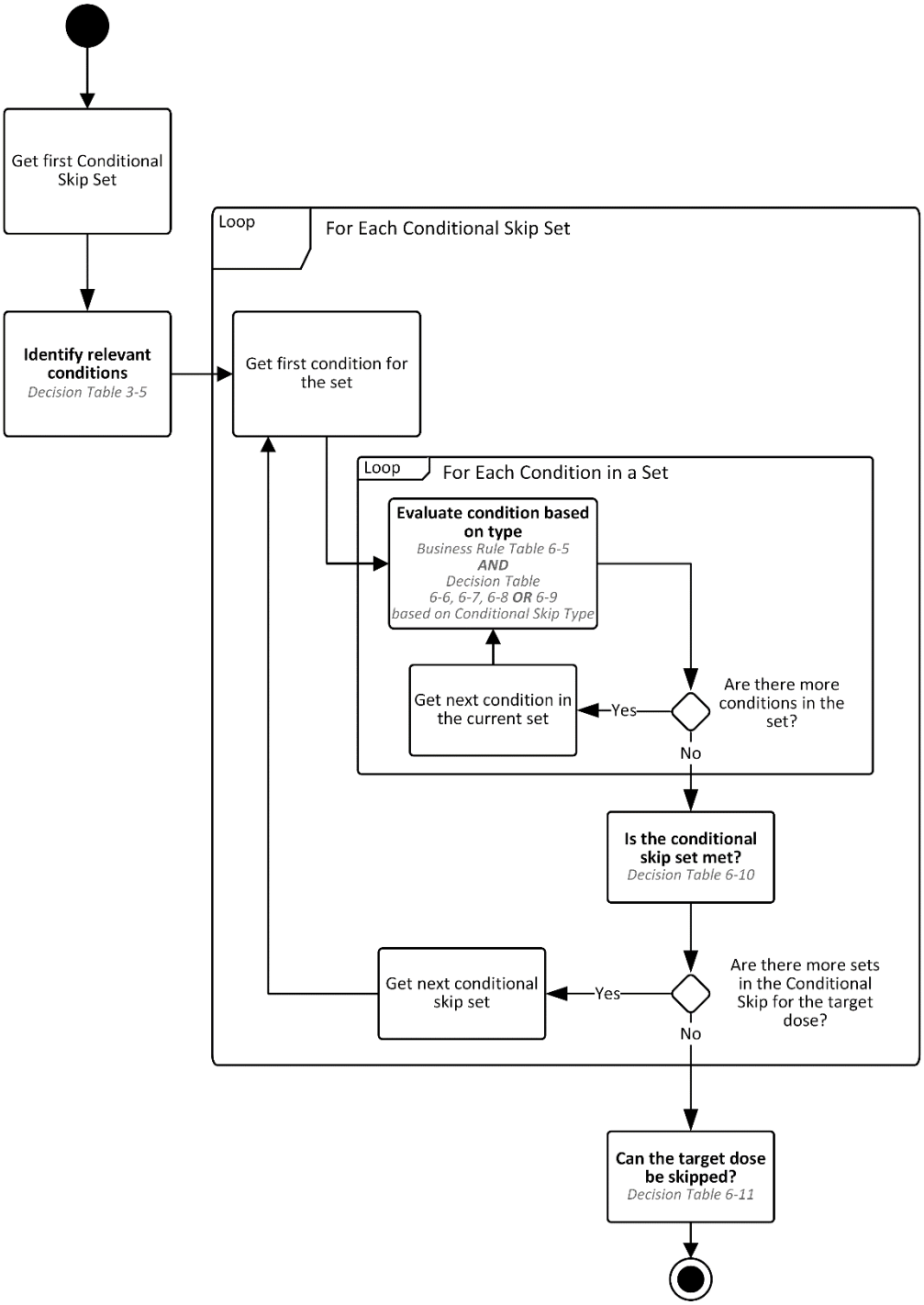

FIGURE 6-3 CONDITIONAL SKIP PROCESS MODEL

TABLE 6-4 CONDITIONAL SKIP ATTRIBUTES

Attribute Type	Attribute Name	Assumed Value if Empty
Vaccine dose administered	Date Administered	-
Patient Immunization History	Administered Dose Count	-
Supporting Data (Conditional Skip)	Conditional Skip elements	-

Attribute Type	Attribute Name	Assumed Value if Empty
Runtime data	Assessment Date	current date
Runtime data	Earliest Date	-
Supporting Data (Conditional Skip)	Start Date	01/01/1900
Supporting Data (Conditional Skip)	End Date	12/31/2999
Calculated date (CALCDTSKIP-3)	Conditional Skip Begin Age Date	01/01/1900
Calculated date (CALCDTSKIP-4)	Conditional Skip End Age Date	12/31/2999
Calculated date (CALCDTSKIP-5)	Conditional Skip Interval Date	-

TABLE 6-5 CONDITIONAL SKIP BUSINESS RULES

Business Rule ID	Business Rule
CALCDTSKIP-3	A patient's conditional skip begin age date must be calculated as the patient's date of birth plus the conditional skip begin age of a conditional skip.
CALCDTSKIP-4	A patient's conditional skip end age date must be calculated as the patient's date of birth plus the conditional skip end age of a conditional skip.
CALCDTSKIP-5	A patient's conditional skip interval date must be calculated as the vaccine date administered from the immediate previous vaccine dose administered plus the Interval of the conditional skip condition.

CONDSKIP-1	The Number of Conditional Doses Administered must be calculated as the count of vaccine doses administered where all the following are true:
	• The vaccine type of the vaccine dose administered is one of the conditional skip vaccine types.
	• The date administered is on or after the conditional skip begin age date and before the conditional skip end age date.
	• The date administered is on or after the conditional skip start date and before conditional skip end date.
	• The Evaluation Status must be one of the following:
	  • 'Valid' when the conditional skip dose type is 'Valid'.
	  • Any status when the conditional skip dose type is 'Total'.

CONDSKIP-2	The Conditional Skip Reference Date must be one of the following: • The Date Administered of the vaccine dose administered when evaluating a vaccine dose administered. • The Assessment Date when determining a forecast. • The Earliest Date when validating a forecast.

TABLE 6-6 CONDITIONAL TYPE OF AGE – IS THE CONDITION MET?

CONDITIONS	RULES	
Is the Conditional Skip End Age Date > Conditional Skip Reference Date ≥ Conditional Skip Begin Age Date?	Yes	No
OUTCOMES	Yes. The condition is met.	No. The condition is not met.

TABLE 6-7 CONDITIONAL TYPE OF COMPLETED SERIES – IS THE CONDITION MET?

CONDITIONS	RULES	
Does the Does the Conditional Skip Series Group identify a Series Group with at least one relevant patient series with a patient series status of 'Complete'?	Yes	No
OUTCOMES	Yes. The condition is met.	No. The condition is not met.

TABLE 6-8 CONDITIONAL TYPE OF INTERVAL – IS THE CONDITION MET?

CONDITIONS	RULES		
Has at least one dose been administered to the patient?	Yes	Yes	No
Is the Conditional Skip Reference Date ≥ Conditional Skip Interval Date?	Yes	No	-
OUTCOMES	Yes. The condition is met.	No. The condition is not met.	No. The condition is not met.

TABLE 6-9 CONDITIONAL TYPE OF VACCINE COUNT BY AGE AND/OR DATE – IS THE CONDITION MET?

Number of conditional doses administered (BR: CONDSKIP-1) / Dose Count Logic	Greater than Conditional Skip Dose Count	Equal to Conditional Skip Dose Count	Less than Conditional Skip Dose Count
Greater Than	Yes. The condition is met.	No. The condition is not met.	No. The condition is not met.
Equal	No. The condition is not met.	Yes. The condition is met.	No. The condition is not met.
Less Than	No. The condition is not met.	No. The condition is not met.	Yes. The condition is met.

TABLE 6-10 IS THE CONDITIONAL SKIP SET MET?

How many conditions were met? / Condition Logic Type	All	At least one, but not all	None
AND	Yes. The set is met.	No. The set is not met.	No. The set is not met.
OR	Yes. The set is met.	Yes. The set is met.	No. The set is not met.

TABLE 6-11 CAN THE TARGET DOSE BE SKIPPED?

How many sets were met?/ Set Logic Type	All	At least one, but not all	None
AND	Yes. The target dose can be skipped. The target dose status is 'Skipped'.	No. The target dose cannot be skipped.	No. The target dose cannot be skipped.

How many sets were met?/ Set Logic Type	All	At least one, but not all	None
OR	Yes. The target dose can be skipped. The target dose status is 'Skipped'.	Yes. The target dose can be skipped. The target dose status is 'Skipped'.	No. The target dose cannot be skipped.

6.3 Evaluate for Inadvertent Vaccine

Evaluate for inadvertent vaccine determines if the vaccine type of a vaccine dose administered was an inadvertent administration due to the vaccine type that was administered.

The following process model, attribute table, Business Rule table and decision table, are used to evaluate for an unallowable vaccine.

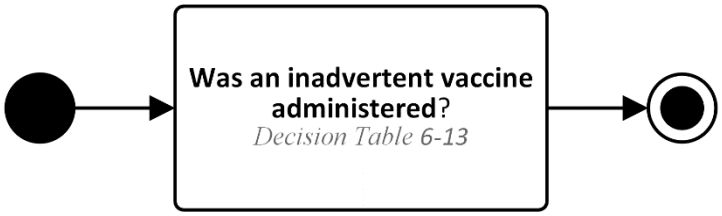

FIGURE 6-4 EVALUATE FOR AN INADVERTENT VACCINE PROCESS MODEL

TABLE 6-12 INADVERTENT VACCINE ATTRIBUTES

Attribute Type	Attribute Name	Assumed Value if Empty
Vaccine dose administered	Vaccine Type	-
Supporting Data (inadvertent Vaccine)	Vaccine Type	-

TABLE 6-13 WAS THE VACCINE DOSE ADMINISTERED AN INADVERTENT ADMINISTRATION FOR THE TARGET DOSE?

CONDITIONS	RULES	
Is the vaccine type of the vaccine dose administered one of the vaccine types of an inadvertent vaccine for the target dose?	Yes	No
OUTCOMES	Yes. The vaccine dose administered was an inadvertent administration for the target dose. Target Dose Status is 'Not Satisfied'. Evaluation Status is 'Not Valid'. Evaluation Reason is 'Inadvertent Administration'.	No. The vaccine dose administered was not an inadvertent administration for the target dose.

6.4 Evaluate Age